Food Consulting Report

Introduction

Our goal in this report is to provide you, the financial advisers, with the most accurate yet simple model to analyze the relationship between household income and spending on eating out. While this analysis may seem straightforward, it is important to note that if the assumptions of a model are not met, the results may be misleading or invalid. This report aims to help you understand this relationship thoroughly, evaluate whether the data supports the hypothesis of a healthy restaurant economy, and provide predictions for how spending may change in the future.

Main Conclusions

Our objective was to determine how household income influences weekly restaurant spending and whether the observed relationship meets the advisers' benchmark for a healthy restaurant economy. Using a heteroskedastic linear regression model, or HLM, which allows both average spending and spending variability to change with income, we find a clear and economically meaningful relationship between income and expenditures on food away from home.

The results indicate that restaurant spending increases steadily as income rises. Specifically, for every additional $1,000 in annual household income, weekly spending on food not prepared at home increases by approximately $0.44. The 95 percent confidence interval for this increase ranges from $0.416 to $0.470, indicating that the positive relationship between income and restaurant spending is strong and precisely estimated. In practical terms, households with greater purchasing power consistently allocate more money each week toward eating out. However, although the effect is statistically strong, its magnitude falls short of the advisers' benchmark of $0.50 per $1,000 increase in income. Because the entire confidence interval lies below $0.50, the data provide clear evidence that the current rate at which restaurant spending rises with income does not meet the advisers' definition of a healthy restaurant economy.

Beyond changes in average spending, the model also reveals that spending variability increases systematically with income. The variance component of the HLM is modeled on the log scale, and the estimated coefficient on income is 0.0136, with a 95 percent confidence interval ranging from 0.0112 to 0.0159. Because this interval lies entirely above zero, we have strong evidence that variability in restaurant spending increases as income increases. Translating this estimate to the original scale, each additional $1,000 in annual income increases the standard deviation of weekly

restaurant spending by approximately 1.37 percent. In other words, as household income rises, not only does expected restaurant spending increase, but the dispersion of spending widens proportionally. Higher-income households therefore contribute more to overall restaurant demand while also introducing greater uncertainty into that demand. From a financial planning perspective, this means that growth in higher-income markets is likely to be accompanied by increased volatility in revenue forecasts.

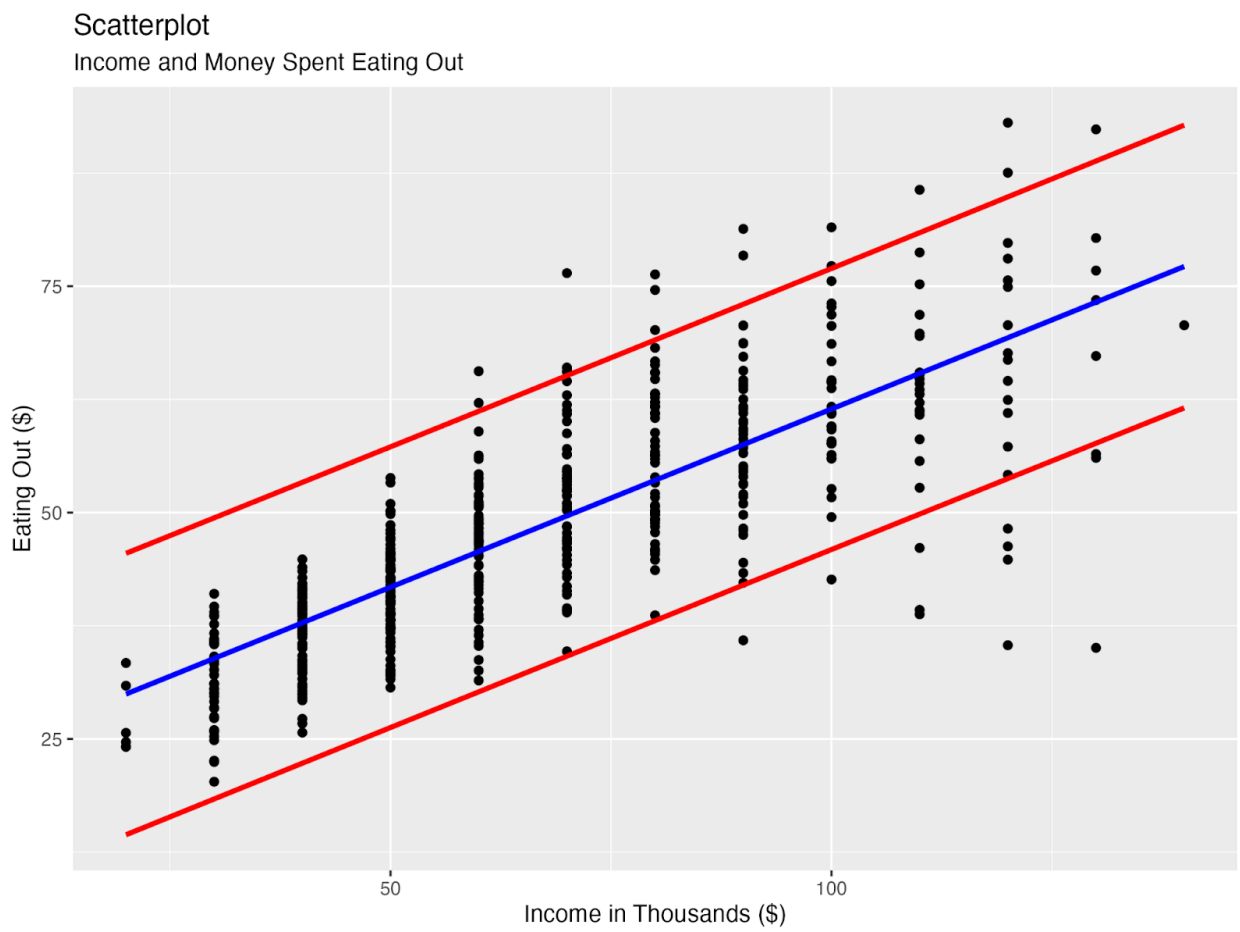
Finally, we evaluated the impact of projected income growth over the next five years, assuming average annual increases of approximately 3.5 percent. We examined households at three representative income levels: $45,000 for the first quartile, $60,000 for the median, and $80,000 for the third quartile. At current income levels, households earning $45,000 are predicted to spend roughly between $29 and $49 per week on restaurants. Median-income households earning $60,000 are predicted to spend approximately between $33 and $58 per week, while households earning $80,000 are predicted to spend between about $38 and $71 per week.

Under projected income growth, these income levels increase to approximately $53,450, $71,260, and $95,010, respectively. With these higher incomes, the predicted spending ranges shift upward in the same order. Households moving from $45,000 to about $53,450 are projected to spend roughly between $31 and $54 per week. Those moving from $60,000 to about $71,260 are projected to spend approximately between $36 and $65 per week. Finally, households whose incomes rise from $80,000 to about $95,010 are projected to spend between roughly $41 and $81 per week. Notably, the width of the predicted spending range increases as income rises. At $45,000, the current range spans roughly $20 per week, compared to about $33 per week at $80,000. After five years of income growth, this spread expands further, reaching nearly $40 per week for higher-income households. This widening gap reinforces the earlier finding that as income increases, restaurant demand not only rises in level but also becomes more variable.

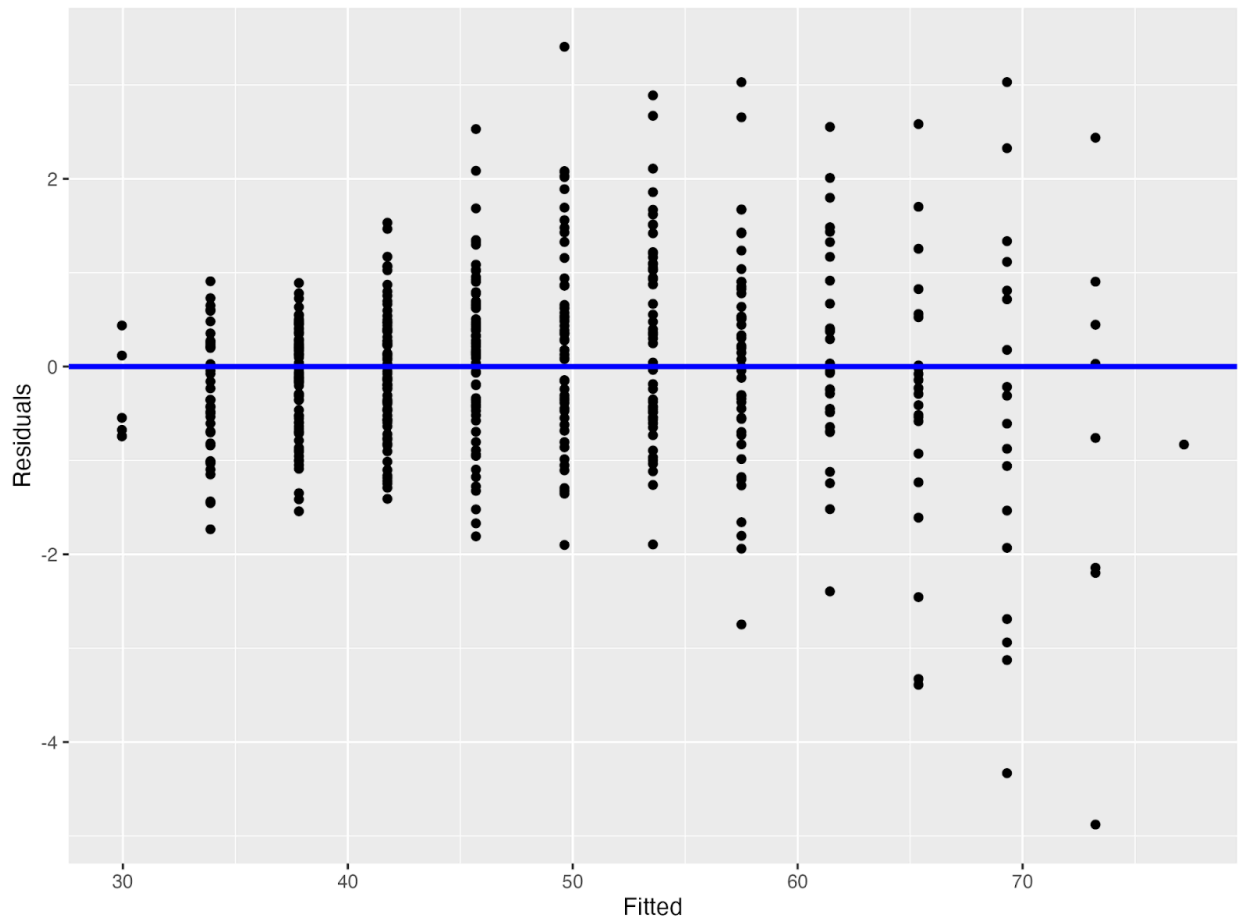
Overall, the evidence suggests that restaurant spending increases consistently with income and will continue to rise as incomes grow. However, the rate of increase, approximately $0.44 per $1,000, falls below the advisers' $0.50 benchmark for a healthy economy. Additionally, spending becomes more variable as income increases, meaning that while higher-income households drive demand growth, they also introduce greater unpredictability. These findings suggest steady expansion in restaurant demand as incomes rise, but at a pace slightly below the benchmark target and with increasing dispersion that should be incorporated into strategic financial planning.

Modeling Details

Now, you may wonder whether we could have simply used a standard linear regression model. In order for such a model to be valid, several assumptions must hold: a linear relationship between explanatory and response variables, independence of observations, normally distributed residuals, and constant variance of the residuals. In our dataset, the primary issue is that the equal variance assumption does not hold.



First, consider the scatterplot showing a standard regression line with a 95% prediction interval. At lower income levels, the interval captures most of the data, but as income increases, it covers fewer observations. This suggests that the variability in spending on eating out grows with income, indicating a change in variance.



Second, the fitted vs. residuals plot provides another perspective. Here, we see a clear “funneling” pattern: the residuals spread further from the horizontal reference line at 0 as the fitted values increase. Ideally, the spread would remain roughly constant, but in this case, the widening pattern highlights unequal variance.

Third, we conducted a formal test for equal variance using the Breusch–Pagan (BP) test. The null hypothesis assumes constant variance, and the alternative assumes variance changes with predictors. Our test returned a p-value near zero, confirming that the residuals do not have equal variance.

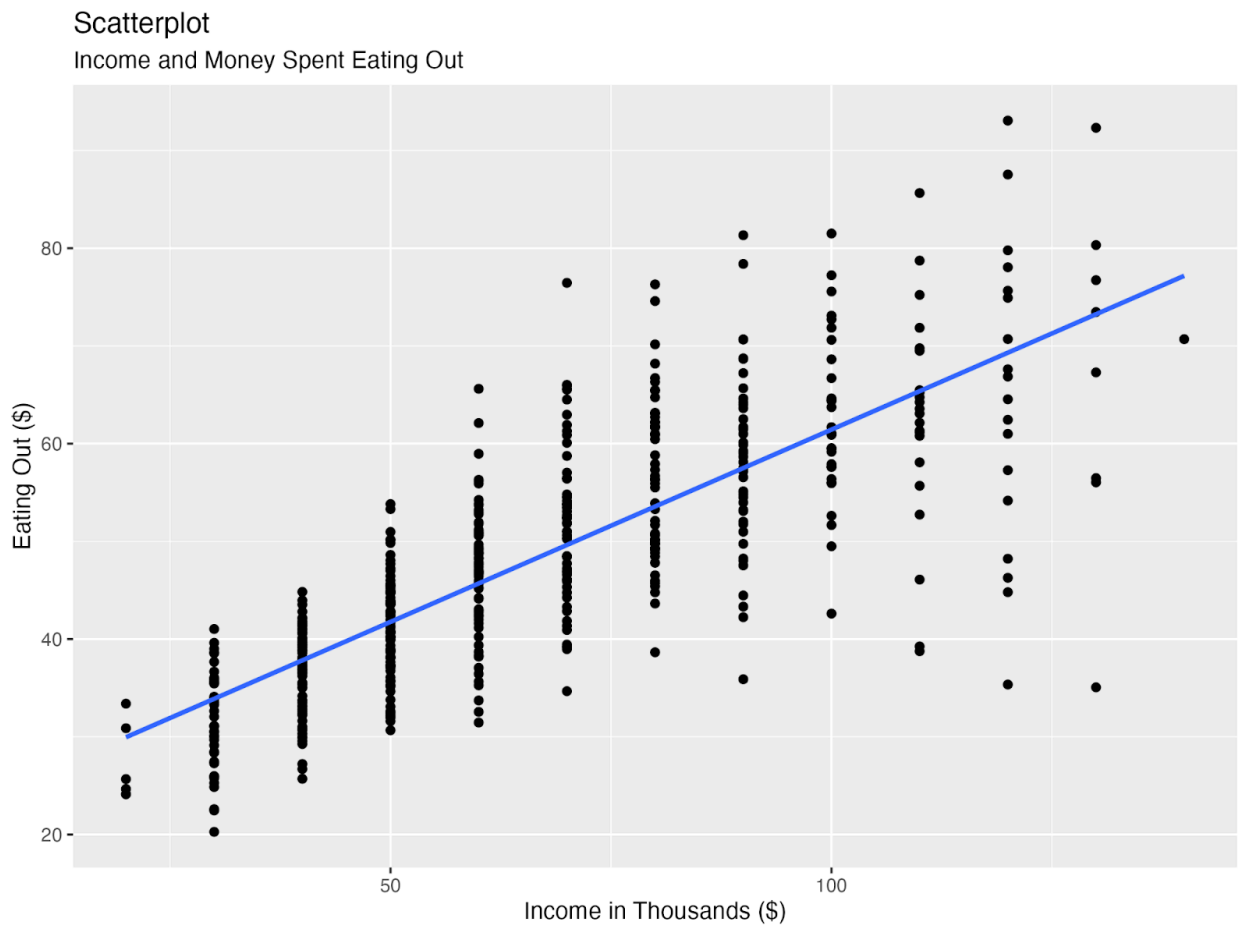
While a standard regression model would still run, ignoring this unequal variance would distort uncertainty estimates. Confidence intervals and hypothesis tests would be unreliable, especially for higher-income households. Because of this, a more advanced modeling approach was necessary — one that accounts for changing variance — which justifies why the National Restaurant Association engaged our expertise and demonstrates why specialized modeling is critical for reliable insights.

Now that we know a standard linear regression model would not work, you may wonder what model we used to account for unequal variance. The model is very similar to a simple linear regression and can be described as follows:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i \text{ where } \varepsilon_i \sim N(0, e^{2(\theta_0 + \theta_1 x_i)})$$

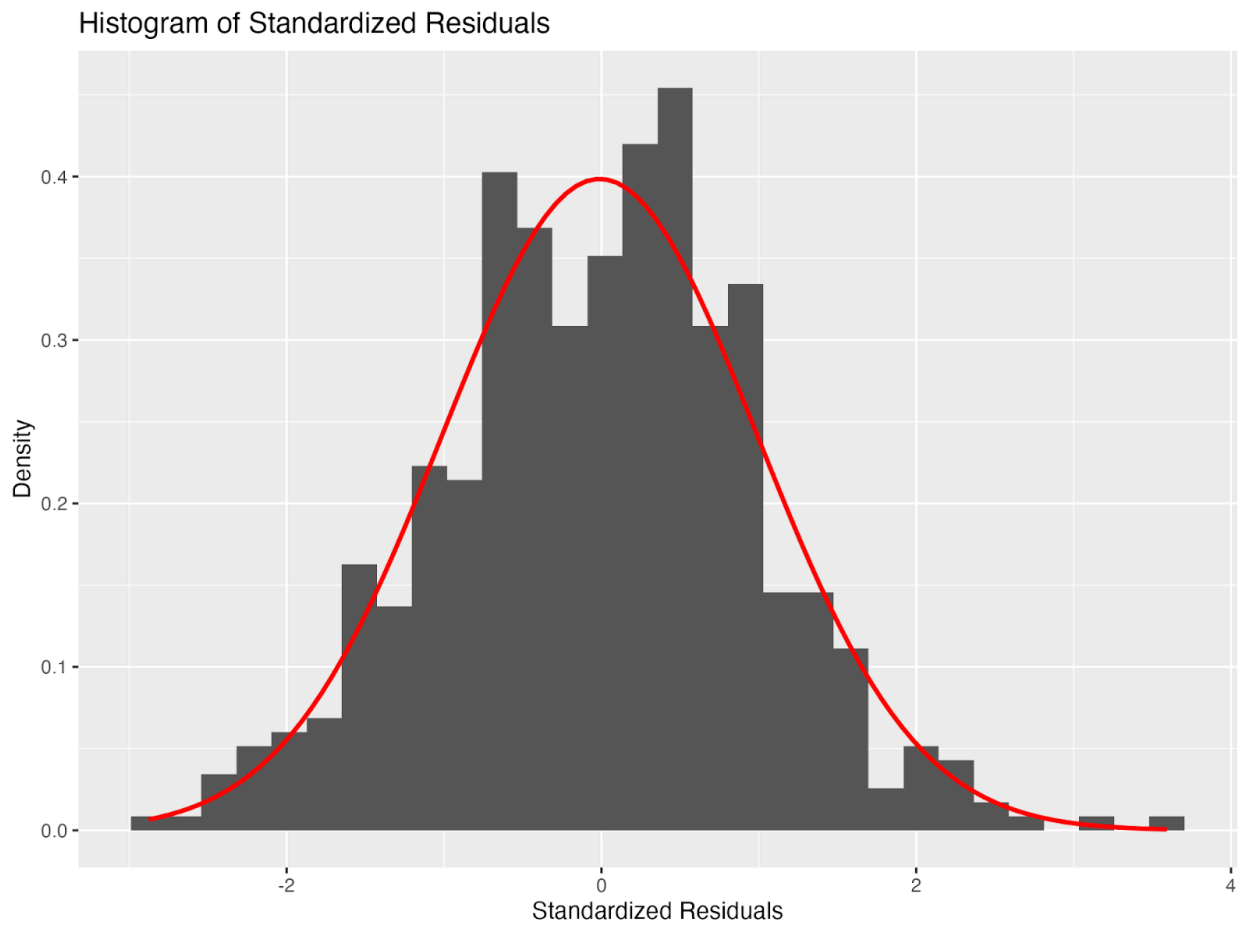
Here, for a given household i, y_i is the weekly spending on eating out, x_i is the income, and ε_i is the prediction error. The key feature of this model is that the variance of ε_i increases with income. Specifically, our formula models the variance as multiplicatively increasing with income, which allows the model to properly account for the wider spread in spending among higher-income households. Using this model, we were able to make predictions and calculate intervals just like in standard linear regression, but now the uncertainty estimates properly reflect the increasing spread of spending as income rises.

To assure you that this model works, we can check the same assumptions we would check with a simple linear regression to make sure our analysis is valid. Specifically, we want to verify a linear relationship between explanatory and response variables, independence of observations, normally distributed residuals, and appropriate handling of variance.

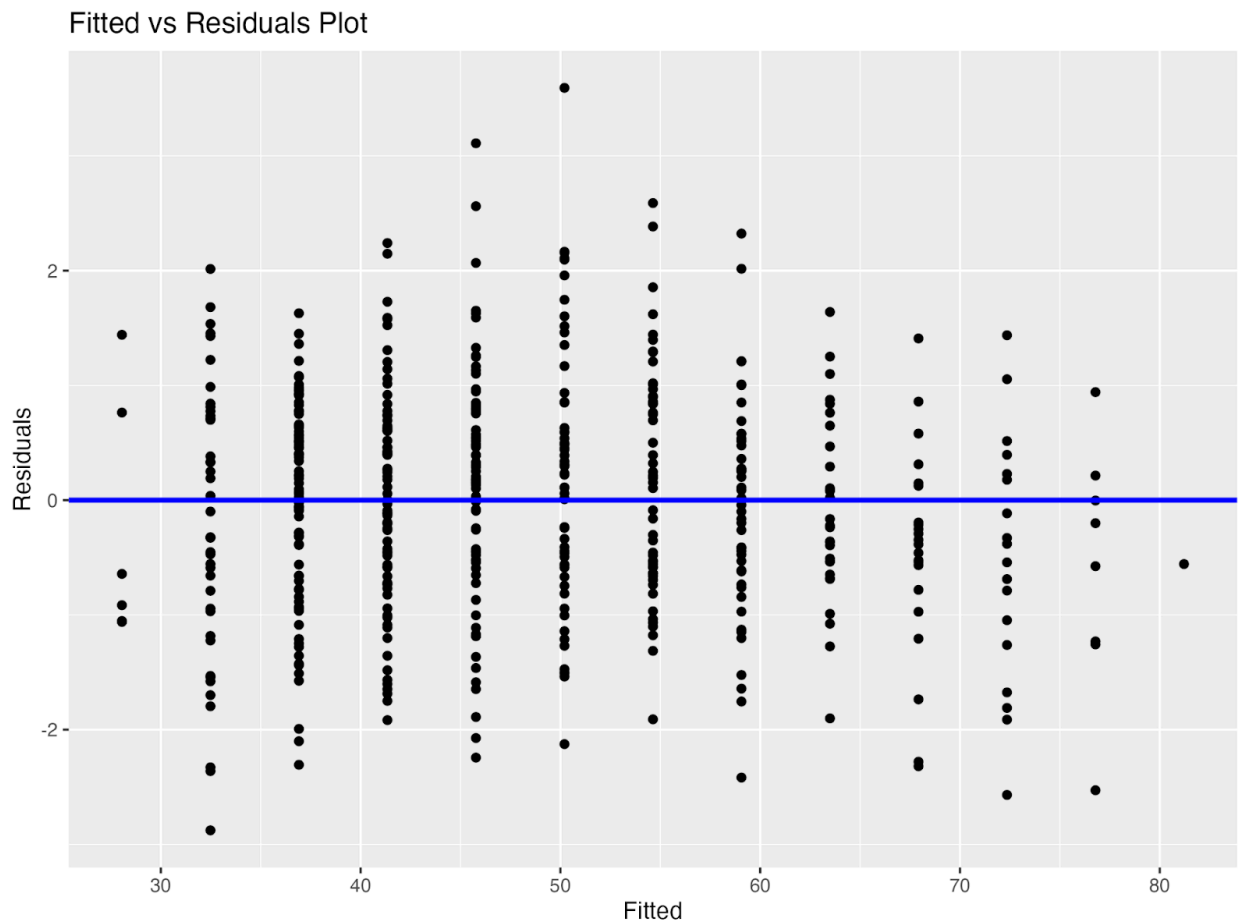


Linearity: Starting with linearity, we can use the same plot as before. The general trend shows that spending on food away from home increases as income increases. This confirms that the linearity assumption is met for our model.

Independence: For independence, we consider how the data were collected. Based on the dataset, there is no reason to believe that the spending of one household would influence the spending of another household. Therefore, we conclude that the independence assumption is reasonable.



Normality: Looking at normality, the residuals appear to follow a normal distribution. To confirm, we ran a Kolmogorov–Smirnov (KS) test. The null hypothesis assumes normality, and the alternative assumes the distribution is not normal. The test returned a p-value of 0.81, which indicates that the residuals are consistent with a normal distribution.



Equal Variance: Finally, regarding variance, our fitted vs. residuals plot now shows a fairly even spread around the horizontal reference line at 0. This indicates that, after adjusting for the unequal variance, the residuals are distributed more evenly. We confirmed this with a Breusch–Pagan (BP) test, which gave a p-value of 0.631, suggesting that the model now meets the equal variance assumption.